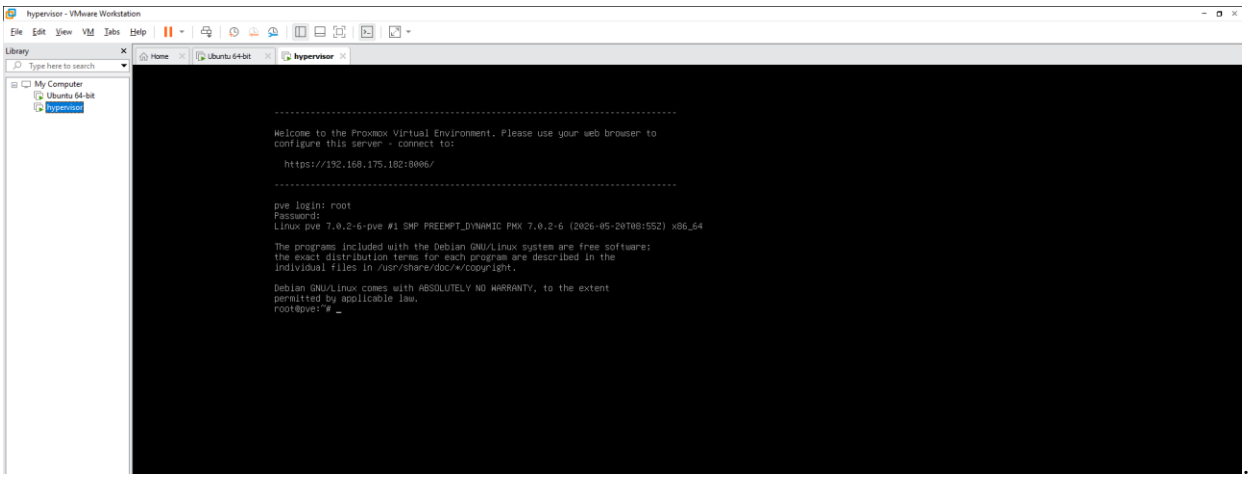
UIT2702 Cloud and Distributed Computing
Exercise 7: Hypervisor Clustering in a Cloud Environment

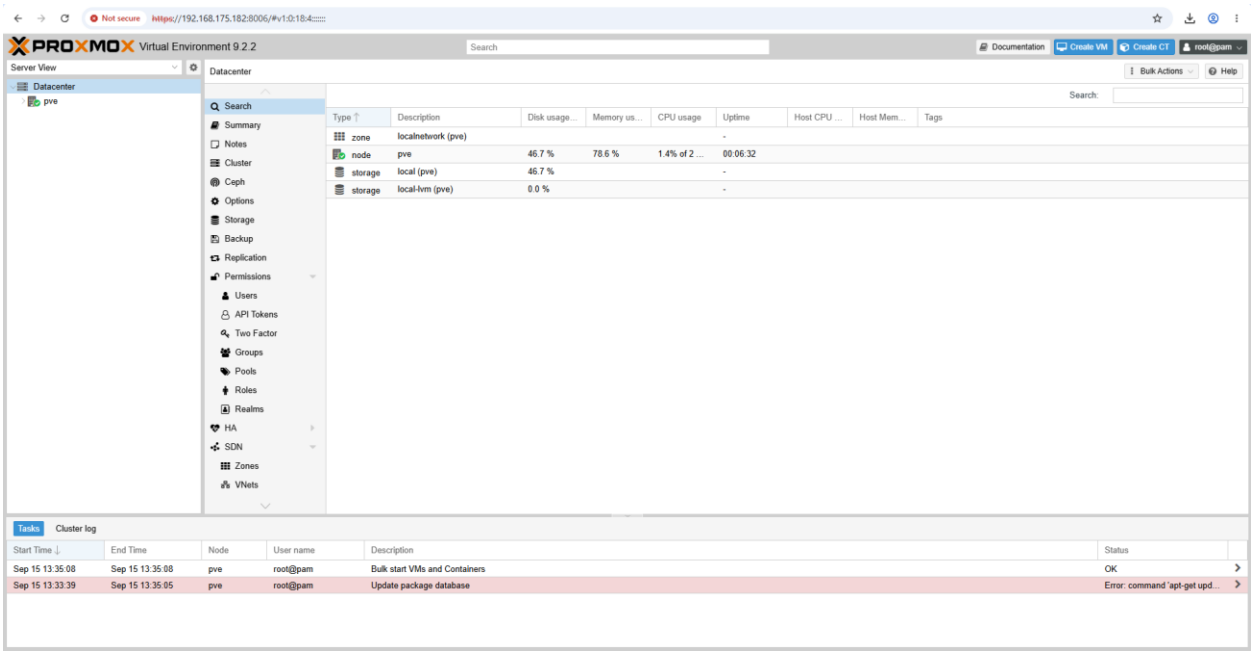
Aim:
To build a nested virtualized hypervisor cluster using VMware ESXi and VMware vCenter Server within VMware Workstation to demonstrate host clustering, High Availability (HA), Distributed Resource Scheduler (DRS) load balancing, and vMotion live migration.

Procedure:
Part 1: Setting Up a Hypervisor Cluster
Task 1: Set Up Hypervisor Hosts
1. Create Virtual Machines for Each Hypervisor Host:
- In VMware Workstation, click File > New Virtual Machine.
- Follow the wizard to create two or more virtual machines that will act as hypervisor hosts. Install VMware ESXi, Proxmox VE, or another hypervisor-compatible OS on each.
- Assign appropriate resources (e.g., 2 CPUs, 8 GB RAM) to each hypervisor host to simulate a realistic cluster.

2. Configure Network Settings for VM Communication:
- Set each hypervisor VM's network adapter to Bridged mode so that each VM can communicate on the same network as the host machine.

3. Install and Configure the Hypervisors:
- Install Proxmox on each VM.
- After installation, access each hypervisor VM through a web interface or console and complete any additional setup (e.g., IP configuration, root password)



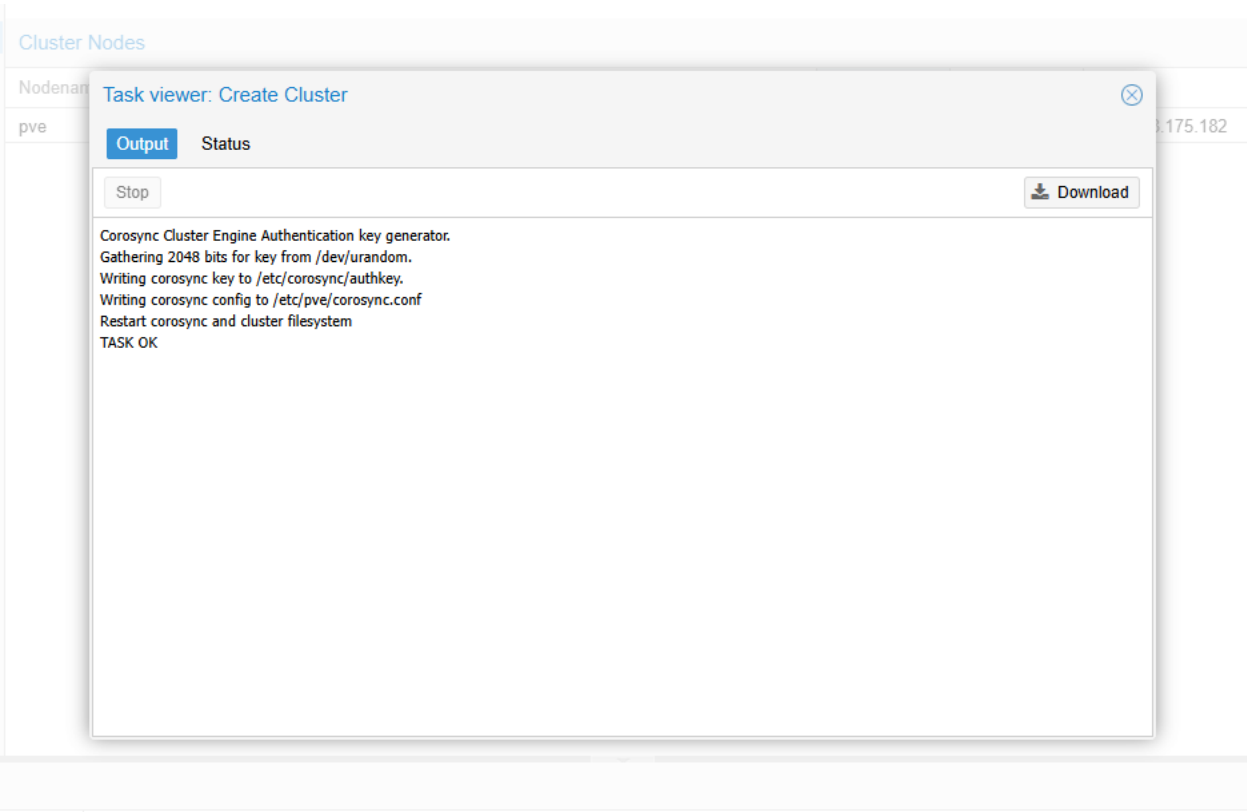


Task 2: Configure Hypervisor Clustering

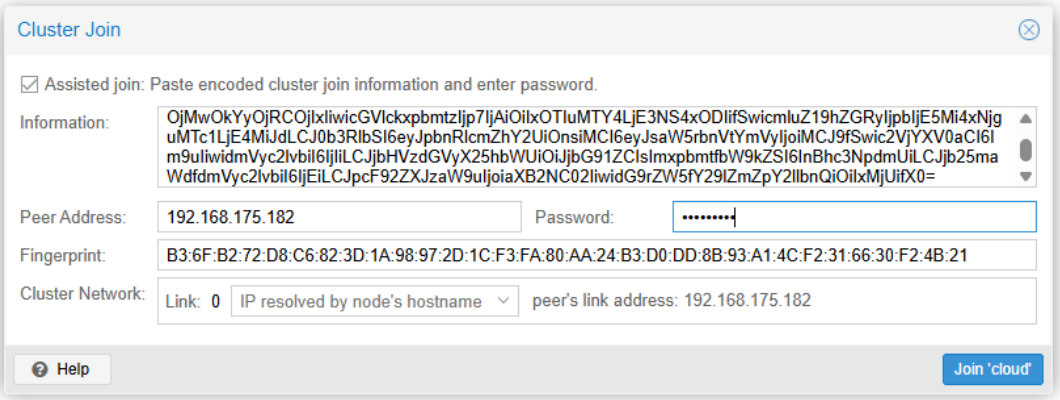
1. Enable Cluster Features on Each Hypervisor:

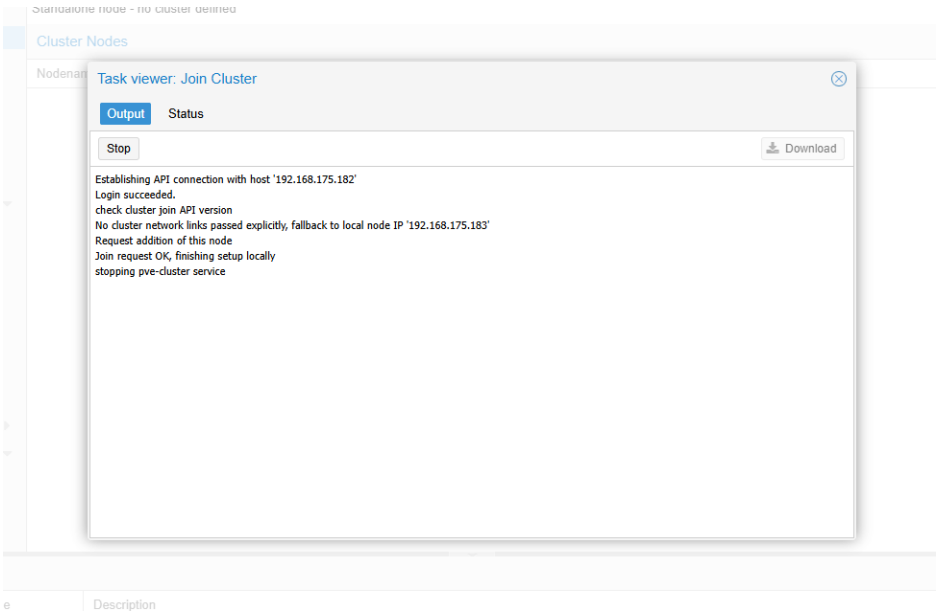
- On each hypervisor (ESXi, Proxmox, etc.), enable clustering and high availability features, if available. For Proxmox:
- In the Proxmox web interface, navigate to Datacenter > Cluster > Create Cluster on the first hypervisor and note the cluster join information.
- On the second hypervisor, join the cluster using the details from the first hypervisor.





2. Verify Cluster Setup:
   - Confirm that each hypervisor node can see the other nodes in the cluster.
   - Test network connectivity between the nodes using ping commands to ensure they can communicate.

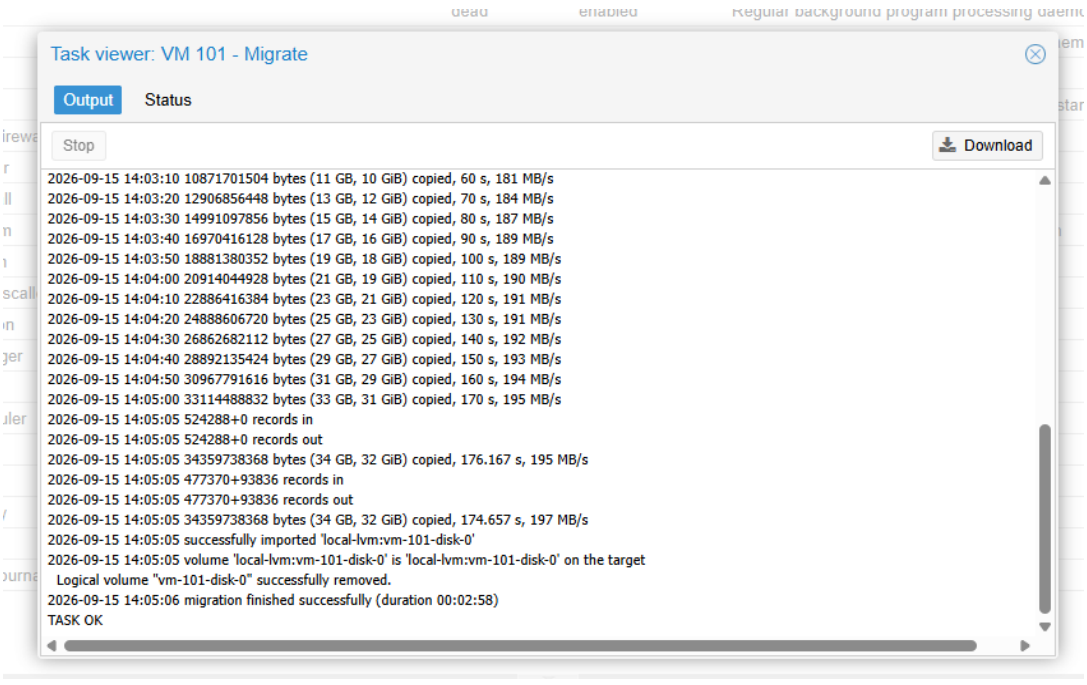




Part 2: Implement High Availability and Load Balancing

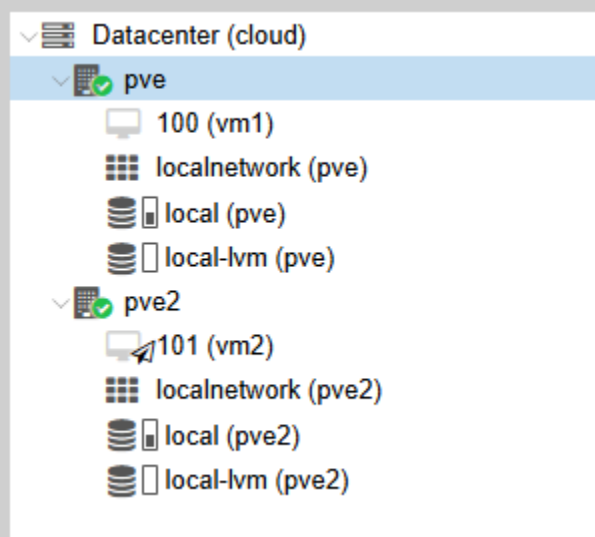
Task 1: Configure High Availability for VMs

1. Enable High Availability in Cluster Settings:
   - In the hypervisor's management interface, navigate to the High Availability settings for your cluster.
   - Specify which VMs should be automatically restarted on another hypervisor in the event of a host failure.
2. Test High Availability:
   - Power down one of the hypervisor hosts and verify that the VM configured with high availability is automatically restarted on another hypervisor.



Task 2: Implement Load Balancing

1. Configure Load Balancing Policies:
   - In the hypervisor's settings, configure resource distribution policies to balance the CPU and memory usage across the cluster.
   - Some platforms allow setting thresholds to move VMs to other hosts when resource usage exceeds defined limits.
2. Monitor and Test Load Balancing:
   - Observe the distribution of VM workloads in your hypervisor's dashboard.
   - Optionally, run CPU and memory-intensive tasks on VMs to simulate load and check if the system balances the load.

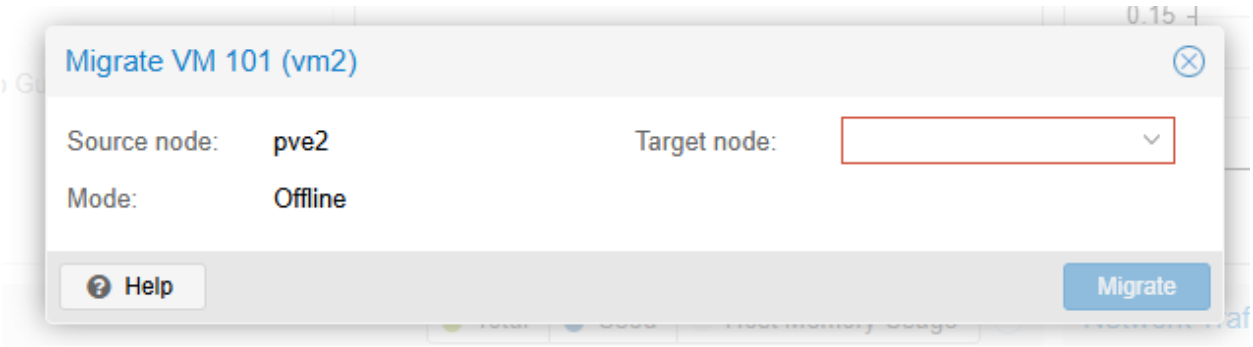


Part 3: Manage and Migrate Virtual Machines

Task 1: Migrate VMs Across Hypervisors

1. Prepare for Migration:
   - Ensure the VMs to be migrated are configured for shared storage and networking so they can operate across multiple hosts.
   - VMware Workstation does not support live migration by default, but you can power off a VM and move it manually between hosts as a simulation.

2. Perform Manual Migration:
   - Power off the VM, then export it as an OVF template or copy its virtual disk files to another hypervisor node.
   - Import the VM on the new hypervisor host and start it up to complete the migration process.
3. Verify Migration:
   - After starting the VM on the new host, verify that it's functioning as expected and connected to the same network as before.



Result:
Hypervisor clustering has been successfully implemented in VMWare Workstation Pro.